

Opinion

Rethinking evidence for epigenetic inheritance in human research

Solveig Løkhammer^{1,*}, Charlotte A.M. Cecil^{2,3,#}, Astanand Jugessur^{1,4,#},
Stéphanie Le Hellard^{5,#}, and Isabelle M. Mansuy^{6,7,#}

In multicellular organisms, epigenetic inheritance is a process by which traits induced by life experiences or environmental factors in a parent are passed to offspring without involving changes in the DNA sequence, but rather in the epigenome. While epigenetic inheritance is evident in many species, from plants to rodents, it remains unproven in humans because of biological barriers, methodological complexity, and the difficulty of disentangling environmental from social and genetic factors. The unique human context of prolonged parental care, as well as cross-generational cultural and societal effects, compounds these difficulties and highlights the need for clear criteria to assess epigenetic inheritance claims. This article aims to help the field move from cataloging biological markers to conducting rigorous human research by clarifying key concepts and defining testable criteria as a guiding framework.

Epigenetic inheritance in humans: challenges and frontiers

Both collective and individual memories and behaviors can be transmitted across generations in humans through culture, social norms, and societal structures [1]. Experimental studies in model organisms, from plants to mammals, indicate that changes in phenotypic traits acquired through experience or environmental exposure can be heritable and passed from the exposed individual to that individual's offspring without any changes in the DNA sequence (as opposed to genetic inheritance) [2]. This challenges the classical view of heredity as the exclusive transfer of information encoded in genetic sequences [3]. This conceptual shift has broad implications for biology, population health, and evolution [4]. However, unlike genetic heredity, which follows universal and empirically testable rules, epigenetic inheritance is mechanistically complex, varies across species, and lacks a unified molecular framework. Testing whether such transmission occurs in humans is challenging, with evidence to date remaining correlational.

Epigenetics encompasses a range of factors and mechanisms, including DNA methylation (DNAm), histone post-translational modifications (PTMs), and RNA-mediated processes that collectively modulate chromatin states and genome activity [5]. These complex regulatory processes confer a high degree of plasticity on genome regulation and enable fine-tuned features such as cellular identity and functional responses to environmental and physiological stimuli. They can be passed from mother to daughter cell during mitotic division (**mitotic epigenetic transmission**; see [Glossary](#)) and from the diploid parental cell to the haploid gamete (**germline**) during meiosis (**meiotic epigenetic transmission**) [6]. At the individual level, **epigenetic inheritance (or transmission)** refers to the transmission of programmed or environmentally induced epigenetic marks from parent to offspring.

When considering epigenetic inheritance in humans, it is essential to take into account some biological features and barriers. First, heritability in this context implies that the marks are present

Highlights

Claims of epigenetic inheritance in humans are overstated as current evidence is correlational and often confounded.

A better empirical framework that considers the principles and biology of epigenetic inheritance would help progress in the field.

Defining stringent, testable, and quantifiable criteria can support more systematic evaluation of evidence.

Understanding potential mechanisms and how to rigorously test and prove them is essential to grasp the difficulty of establishing causality in humans.

¹Centre for Fertility and Health, Norwegian Institute of Public Health, Oslo, Norway

²Department of Child and Adolescent Psychiatry, Erasmus MC, Rotterdam, The Netherlands

³Department of Epidemiology, Erasmus MC, Rotterdam, The Netherlands

⁴Department of Global Public Health and Primary Care, University of Bergen, Bergen, Norway

⁵Department of Clinical Science, University of Bergen, Bergen, Norway

⁶Laboratory of Neuroepigenetics, Medical Faculty, University of Zürich, Zürich, Switzerland

⁷Department of Health Sciences and Technology, ETH Zürich, Zürich, Switzerland

[#]These authors contributed equally.

*Correspondence:
solveig.lokhammer@fhi.no
(S. Løkhammer).

in the germline. While exposure can affect somatic and reproductive cells simultaneously, **epigenetic modifications** in somatic cells can be passed only to other somatic cells through mitosis and thus do not contribute to germline-dependent inheritance. Changes observed in peripheral cells, such as those in blood, do not necessarily reflect epigenetic or RNA states in the germline. For epigenetic heritability, changes need to be maintained until mature gametes form and then passed to the offspring. Although communication between the soma and the germline was long postulated to be unidirectional (Weismann barrier theory), it is now known to be bidirectional in mammals [7]. In rodents, small RNAs or blood components can reach the germline and modify it, with consequences for the offspring [8,9]. Such signal transfer constitutes a possible route for epigenetic inheritance.

Second, for changes to be passed from an exposed parent to offspring, they should resist **epigenetic reprogramming** (Box 1). Reprogramming is a process of epigenetic resetting in the early embryo that erases and then re-establishes 80–90% of parental epigenetic marks [12]. However, many DNA and histone PTMs are maintained, and RNA *per se* is not reprogrammed during development. The constraint of reprogramming is therefore not a strict obstacle to epigenetic inheritance. Indeed, the inheritance of environmentally induced changes has been documented in many phylogenetically and functionally diverse species [13]. This raises the possibility that, in humans, life experiences, especially when extreme and chronic (e.g., domestic violence, trauma, poor nutrition, and exposure to endocrine disruptors), may affect the descendants of an exposed parent.

Given the complexity and partial mechanistic understanding of epigenetic inheritance in mammals, studying this process in humans is highly challenging. To advance the field, it is important to establish a rigorous conceptual framework and distinguish epigenetic inheritance from other forms of information transfer. Some molecular mechanisms have been proposed [14–18], but the aspects and criteria that are testable and measurable in humans need to be more clearly delineated. Current evidence and claims of epigenetic inheritance in humans also need to be carefully re-evaluated. In this opinion article, we discuss these points and aim to promote conceptual and methodological rigor in research in this field.

Current findings on epigenetic inheritance in animal models and humans

Experimental studies in rodents and other model organisms have demonstrated that poor diet, traumatic stress, or endocrine disruptors in one generation can induce phenotypic changes in subsequent generations, even if they are not directly exposed themselves [19–21]. In humans, epidemiological studies have reported consistent phenotypic traits in the descendants of exposed individuals across generations. In the Dutch Hunger Winter cohort, prenatal famine exposure was associated with differences in DNAm measured in adults decades after exposure [22,23]. Furthermore, two Swedish cohorts—Överkalix and a more recent, larger cohort from Uppsala—reported a link between the grandfather's food access before puberty and mortality in grandsons [24,25]. Periconceptional conditions such as season of conception or nutritional intake have also been associated with persistent epigenetic signatures at metastable **epialleles**

Box 1. What is epigenetic reprogramming?

Epigenetic reprogramming in mammals occurs in two waves: the first wave before implantation and the second during PGC specification in developing gonads. In humans, methylome profiling of preimplantation embryos shows that DNA demethylation starts between fertilization and the two-cell stage and is accompanied by rapid remodeling of histone PTMs, with reduced repressive marks such as histone H3 lysine 9 di-/trimethylation (H3K9me2/3) and histone H3 lysine 27 trimethylation (H3K27me3), and increased activating marks such as histone H3 lysine 4 trimethylation (H3K4me3) at key regulatory regions. Later in the embryo, only sparse changes occur until implantation, when *de novo* methylation is re-established. In PGCs, genome-wide CpG methylation progressively declines from levels comparable to surrounding somatic cells (75–80%) to 4–5% by weeks 7–9 of development (~90–95% loss) [10]. Likewise, repressive histone PTMs diminish relative to surrounding somatic cells [11].

Glossary

Biological embedding: process by which environmental factors and life experiences become durably incorporated into biological systems, shaping long-term physiology and health.

Direct exposure: exposure of an individual to an environmental factor or a life event that may result in measurable phenotypic effect(s). The effect(s) can occur through different mechanisms, such as epigenetic modifications in different cells of the body, depending on the type, timing, and duration of exposure, which can modulate signal transduction pathways, and/or through DNA damage or mutation (e.g., ultraviolet radiation, teratogens, and carcinogens).

Direct genetic effects: how an individual's genome influences traits and behaviors.

Epiallele: heritable variant of a locus that differs in its epigenetic state rather than in the underlying DNA sequence. An epiallele can arise stochastically, be driven by genetic variation (e.g., methylation QTLs), and/or be metastable (e.g., established in early development and maintained).

Epigenetic inheritance (or transmission): transfer of epigenetic marks or states that can influence phenotypes without changes in the DNA sequence, whether programmed or environmentally induced, from a mother cell to daughter cells or from a parent to its direct (intergenerational) or subsequent (transgenerational) offspring. In multicellular organisms, the term epigenetic inheritance implies that the germline is involved, resulting in the transfer of epigenetic changes from one generation to the next via gametes without requiring direct exposure in each generation.

Epigenetic modifications: chemical modifications to DNA, histone proteins, or certain RNAs (epitranscriptomic) that for DNA and histones regulate genome activity, including gene expression, without altering the underlying DNA sequence and for RNA modulate RNA stability and function.

Epigenetic reprogramming: genome-wide partial erasure and re-establishment of epigenetic marks, occurring in two waves in the early embryo to confer totipotency (the capacity of a single cell to form an organism) and developmental potential,

[26,27]. These epialleles are established before cell-lineage commitment and are detectable in many tissues [28]. However, such effects may result from embryonic exposure to maternal factors such as circulatory molecules influenced by seasonal infections or vitamin D levels, which can modify the fetal epigenome [29–31].

Other studies have linked childhood maltreatment, lifestyle (smoking and BMI), or socio-demographic factors (age and socioeconomic status) to epigenetic modifications in sperm, suggesting experience-dependent **biological embedding** in germ cells [32]. Comprehensive profiling of sperm from lean and obese men revealed similar histone positioning but markedly different small noncoding RNA expression and DNAm patterns [33]. In a separate cohort of morbidly obese men, surgery-induced weight loss was also associated with substantial remodeling of sperm DNAm, particularly at genomic regions implicated in appetite regulation [33]. These findings suggest that the epigenome of human spermatozoa is dynamically modulated in relation to obesity and that metabolic dysfunctions may be propagated to subsequent generations. DNAm changes have been detected in cord blood of newborns whose mothers had a history of childhood trauma [34,35], suggesting intergenerational risk transmission. However, these studies are observational and do not distinguish between factors such as *in utero* exposure, re-exposure through parental behaviors or shared environments, and/or genetics. They also do not indicate whether changes in germ cells are transmitted (Box 2) [36,37]. Thus, human studies cannot conclusively determine whether environmentally induced epigenetic modifications are transmitted and cause phenotypic changes in the offspring (Figure 1).

Box 2. Illustrative model of epigenetic inheritance following preconception stress

For the inheritance of environmentally induced epigenetic changes to contribute to human health, certain criteria should be met. Using stress exposure before conception and offspring vulnerability to mental disorders (e.g., stress-related phenotypes) as an example, this section outlines the features of this potential process (Figure 1).

- (i) Exposure definition: stress exposure should be defined as precisely as possible and, when possible, restricted to a specific life period in the parent(s), for example, before the age of 10 or less than 6 months before conception. Longitudinal and negative control designs can strengthen causal inference between parental exposure and offspring outcomes by tracking phenotypic changes over time and comparing exposed to unexposed groups. Importantly, such research designs facilitate the investigation of causal relationships between parental stress exposure and offspring symptoms. Evidence for epigenetic mechanisms requires direct molecular data and should not be inferred solely from phenotypic associations.
- (ii) Exclusion of confounders: the association between parental exposure and offspring stress reactivity should persist after controlling for genetic confounding, intergenerational environmental continuity, and other factors such as socioeconomic status. Family-based comparisons, including full- or half-sibling analyses, address shared familial influences but do not fully account for genotype effects. Incorporating parental and offspring genotype data, as in trio designs, enables further separation of direct and indirect genetic effects. Nevertheless, these approaches may not capture prenatal environmental influences, such as *in utero* exposure. Negative control designs, such as comparisons of maternal and paternal exposures, can help distinguish intrauterine from shared familial stress effects.
- (iii) Germline relevance: exposure-induced epigenetic alterations should be detectable in mature germ cells (e.g., sperm), rather than exclusively in somatic tissues or cells, such as those in blood, thereby supporting plausible inheritance via the germline. Animal and experimental studies can be used to investigate epigenetic changes and their transmission from an exposed parent to its direct offspring and subsequent descendants.
- (iv) Epigenetic factors: epigenetic changes associated with exposure, identified in germ cells, should also be present to some extent in early embryonic or offspring tissues or cells (e.g., blood). Maintenance of epigenetic changes would suggest escape from epigenetic reprogramming. Maintained changes may have functional relevance for stress-related biological pathways, directly or via downstream molecular components (e.g., transcription factors and their gene targets). Integrating germline and offspring molecular data, with supporting evidence from animal models, can strengthen this inference.
- (v) Functional consequences: offspring should exhibit reproducible differences in 'stress reactivity'—a transdiagnostic trait closely associated with psychiatric risk. Ideally assessed prospectively, consistent associations between stress-related molecular alterations (e.g., higher blood cortisol or proinflammatory cytokines) and phenotypes (e.g., psychiatric symptoms) in the offspring, together with the recurrence of those phenotypes, would corroborate parent-to-offspring effects.

with selective preservation at imprinted genes, repetitive elements, and other discrete regions.

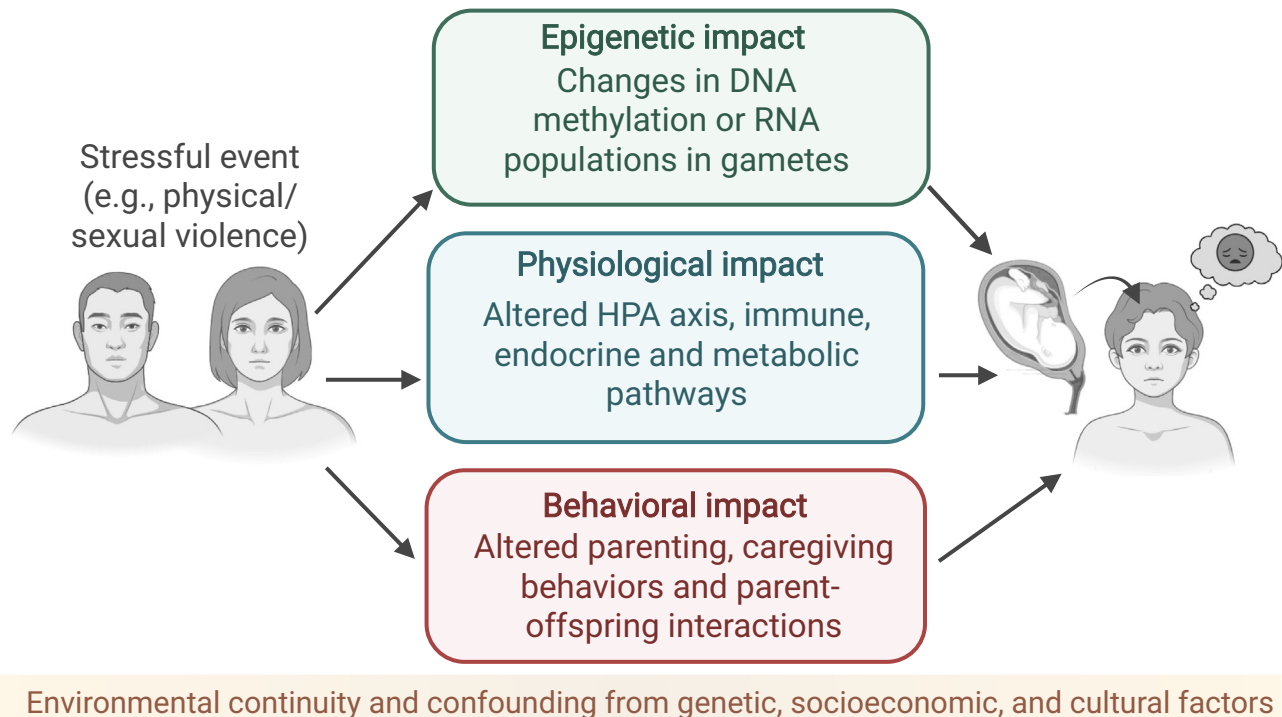
Genetic inheritance (or transmission): transfer of genetic information, including DNA sequence variants, chromosomal structures, and mitochondrial DNA, from parents to offspring via gametes for sexually reproducing species and via direct cellular replication of the genome (e.g., mitosis, fission, and budding) for asexually reproducing organisms.

Germline: lineage of reproductive cells that gives rise to gametes (oocytes and spermatozoa). Gametes, unlike somatic cells, are haploid and carry one copy of each chromosome (half of the parental genetic material).

Indirect genetic effects: effects of an individual's genes on the phenotypes of another individual, mediated through the environment one creates for the other individual (e.g., via physiological, developmental, or behavioral pathways) rather than by direct genetic transmission (e.g., parental genes influencing offspring outcomes even when these genes are not passed on).

Meiotic epigenetic transmission: passage of epigenetic states through meiosis into gametes, leading to the transfer of epigenetic information from parent to offspring.

Mitotic epigenetic transmission: stable propagation of epigenetic marks or states during somatic cell division within an organism, ensuring the maintenance of parental cell identity in daughter cells.



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Figure 1. Challenges in inferring epigenetic inheritance from parental exposure and offspring phenotypes, illustrated using preconceptional stress and psychiatric outcomes. Genetic variants and socioenvironmental factors can shape phenotypes across generations to different degrees and are difficult to disentangle in humans. Epigenetic mechanisms lie at the intersection of these factors and can respond to them dynamically. Pathways may differ for maternal versus paternal stress exposures. Although the schematic shows only one generational step for clarity, the requirements for distinguishing direct, intergenerational, and transgenerational epigenetic effects are detailed in [Figure 2](#). This figure was created using BioRender. Lokhammer, S. (2026). HPA: hypothalamic–pituitary–adrenal.

Overcoming biological barriers

Biological inheritance is traditionally viewed as the transmission of genetic material in the form of the DNA sequence from parent to offspring via the germline [**genetic inheritance (or transmission)**]. The DNA sequence is largely insulated from environmental influences and is rather stable, except in the case of exposure to ultraviolet radiation, teratogens, or carcinogens that can cause genetic mutations. In contrast, epigenetic inheritance is the transfer of information beyond the DNA sequence, which includes biochemical modifications of the DNA itself and its associated proteins (histones), as well as RNA. These factors are dynamically modulated by the environment and life experiences in all cells of the body, in some cases persistently. They can alter genome activity, including gene expression, and thereby participate in phenotypic changes. When present in gametes, they can be passed to the embryo at fertilization and, if maintained across reprogramming and development, reproduce phenotypic changes in the next generation [38]. Maternal and paternal germlines have different developmental trajectories. Most oocyte development occurs prenatally, after which oocytes remain arrested for extended periods [39]. In contrast, spermatogenesis is a continuous process throughout adult life, characterized by repeated rounds of cell division and ongoing environmental susceptibility [40]. These distinct developmental timelines create different temporal windows during which environmental exposure may influence germ cells, potentially contributing to differences in the timing, persistence, and mechanisms of environmentally induced epigenetic changes [40].

In the early embryo, the parental epigenome is partly remodeled to establish a chromatin ground state permissive for development (Box 1). Despite reprogramming of DNAm and histone PTMs, residual cytosine-phosphate-guanine (CpG) methylation and repressive histone marks are maintained at multiple loci in human primordial germ cells (PGCs). These include parent-specific imprinting control regions, some developmental genes, subsets of evolutionarily young transposable elements, and other potentially deleterious genomic sequences [38]. Some histone PTMs are also retained, but more broadly than only at the restricted set of protected loci. Reprogramming in humans has a different temporal window compared with that in mice, in which global demethylation already occurs in the zygote [41]. The kinetics of DNA demethylation in early human embryos also differ between maternal and paternal genomes, with maternal demethylation being more gradual than paternal demethylation [42]. Although DNA demethylation kinetics differ between parental genomes, both undergo extensive reprogramming of histone PTMs during early embryogenesis [42].

For epigenetic inheritance, parental temporal differences may influence the consequences of exposure on the embryo. Maternal high-fat diet affects the oocyte epigenome and small RNA content and progressively alters imprinting and gene regulation during pre- and post-implantation development [43]. In contrast, paternal high-fat diet modifies sperm small RNA cargo, particularly tRNA-derived fragments that are delivered to the zygote, and alters gene expression in the preimplantation embryo [44]. This is consistent with the rapid versus slow reprogramming kinetics of the maternal and paternal genomes.

RNA has also been implicated in the transfer of information from parent to offspring. Unlike DNA or histone modifications, RNA molecules are not reprogrammed in the sense of enzymatic erasure and re-establishment. Oocytes are extraordinarily rich in maternal RNAs, which are accumulated during oogenesis. Upon fertilization, sperm-delivered RNAs are introduced into the oocyte and coexist transiently with maternal RNAs until the early two- to eight-cell stages of the embryo [45,46]. Gametic RNAs in the embryo undergo progressive decay. Maternal transcripts are actively deadenylated and degraded during the maternal-to-zygotic transition, and most are cleared by the eight-cell stage in humans (two-cell stage in mice). Sperm-delivered RNAs are also likely diluted and degraded during this window [46,47]. However, sperm RNAs modified by exposure can be detected in two-cell embryos after fertilization and exert functional effects on subsequent developmental stages, up to adulthood. Sperm total RNA from mice exposed to postnatal traumatic stress, when microinjected into control zygotes, is present in the embryo and reproduces behavioral and metabolic phenotypes in the offspring [48]. Likewise, sperm tRNA fragments modified by paternal dexamethasone treatment are persistently altered in two-cell embryos and modify transcriptional trajectories and metabolism in the offspring [49]. These findings suggest that environment-sensitive sperm RNAs can persist and are sufficient to establish durable chromatin states and transcriptional programs in the early embryo.

The fact that epigenetic marks and RNA can be modified by parental exposure and that these modifications can be maintained across reprogramming makes epigenetic inheritance plausible in mammals. The multiplicity and diversity of modifications, their coexistence [50] and their possible combined action can collectively drive gene-regulatory changes that can propagate into adulthood. Such a multilayer view is key because it shifts attention away from any single epigenetic mark or RNA toward the combined influence of multiple inherited epigenetic factors on offspring development and phenotype. This underscores the complexity of epigenetic inheritance, which must be fully acknowledged rather than simplified.

Recognizing its multifactorial and dynamic nature can avoid reductionist interpretations and misleading conclusions and help design appropriate and rigorous studies.

Distinguishing routes of parental influence on offspring phenotypes

Environmental factors influence phenotypes to varying degrees and via different routes across species [51]. Humans are distinguished by high levels of parental care and by the strong impact of cultural and socioeconomic conditions, such as education, on children and grandchildren [52, 53]. The long generational interval and the complex interplay of environmental and social factors make it difficult to distinguish germline-mediated inheritance from **direct exposure** (e.g., *in utero* exposure or behavioral and social influences) [54]. Thus, offspring behavior can be shaped by parental and societal factors, which may also induce changes in epigenetic states and genome regulation. This complexity commonly results in conceptual ambiguity and misinterpretation of findings as evidence of inheritance. For instance, correlated epigenetic changes and phenotypes cannot be claimed to be evidence of epigenetic mediation of the effects of trauma if the repetition of trauma (e.g., through parental behavior) cannot be excluded. Individuals exposed to childhood maltreatment are at increased risk of perpetuating such behavior, resulting in the reiteration of direct exposure being wrongly assigned to intergenerational transmission [55]. Conversely, experiences such as education and interventions like psychotherapy can mitigate the recurrence of trauma [56]. In mice, an enriched environment after postnatal trauma can correct behavioral deficits and protect the offspring [57].

Clarifying the influence of direct and indirect genetic effects on environmentally induced epigenetic modifications

Epigenetic marks are modified depending on the cellular context, intrinsic biological states (age, sex, and hormonal milieu), environmental exposures (smoking, nutrition, and stress), and underlying genetic background. Some epigenetic marks are determined by DNA sequence variants and are thus heritable, representing **direct genetic effects** [58,59]. CpG-creating SNPs can dictate whether a locus is methylated or not, or variants in transcription factor binding sites can recruit or exclude chromatin modifiers, leading to allele-specific histone PTMs. Such sites, often identified as quantitative trait loci (QTLs), can act independently or in interaction with exposures [60]. For example, a given exposure may induce epigenetic modifications and phenotypic changes in some individuals but not others, depending on their genotype [61,62]. Integrating genetic variation into epigenetic analyses is essential to assess causality and clarify whether and how environmental sensitivity itself is genetically encoded. This is also highly relevant for RNA, which is sensitive to exposures and can convey environmental information to offspring via the germline. Besides direct effects, genetic factors can shape behaviors commonly attributed to environmental influence through **indirect genetic effects**.

Indirect genetic effects are a well-characterized type of effect in mammals, in which genetically influenced parental traits, such as gestational physiology, lactation, or postnatal care, shape offspring phenotypes independently of the alleles that are directly passed to the offspring [63]. Because the environment provided by parents and relatives may itself be similar across generations, indirect genetic effects through nontransmitted alleles create non-Mendelian pathways of inheritance in which environmentally mediated variation is passed from one generation to the next [64]. This distinction is pertinent to studies of epigenetic inheritance because traits caused by the developmental environment and parental behavior can phenocopy a trait and therefore be confounded with germline-based epigenetic inheritance [65]. In experimental models, this can be tested by cross-fostering and strategies that isolate germline factors, such as sperm RNA injection, which decouple parental phenotypes from information transmitted by gametes [44,48].

Understanding the distinction between direct and indirect genetic effects is crucial, as both can modulate the way environmental exposure leaves an epigenetic mark across generations. For example, the impact of childhood maltreatment on an individual can be modulated by parental genomes, which can influence transmission to the next generation [66]. Several genetic variants associated with childhood maltreatment have been identified [67]. While environmentally induced epigenetic modifications can be passed from an exposed parent to offspring, the interaction between genetic and epigenetic factors is highly complex. Disentangling these factors is challenging without experimental controls. In animal models, genetic diversity can be minimized by using inbred strains, which helps isolate the effects of exposure itself [68]. This is not achievable in humans, but cohorts with reduced genetic heterogeneity may serve a similar purpose [9].

Ethical challenges of human studies

In humans, experimental settings necessary to demonstrate epigenetic inheritance are inapplicable. In laboratory rodents, parental conditions can be manipulated by stress, diet, or endocrine disruptors and circumscribed to a single generation (e.g., male mice or rats do not participate in raising their offspring and never meet them). In humans, it is possible to assemble cohorts exposed to adverse life events such as domestic violence, war trauma, or poor nutrition, although identifying appropriate unexposed control groups is difficult [9,69]. Adoption studies are also useful to exclude postnatal parental influences [70] but they are rare, cannot rule out prenatal effects, and are subject to biases, such as psychosocial effects. Access to human tissues—particularly embryonic and reproductive cells—is limited and ethically challenging, making epigenetic analyses burdensome [71]. Despite these constraints, human studies remain essential and, to some extent, irreplaceable, as animal models cannot fully recapitulate the complexity of human experiences. Large, longitudinal population cohorts can help partially compensate for such complexity by enabling adjustment for multiple confounding variables and the temporal sequence. However, the link between parental exposure, the germline epigenome, and offspring phenotype will remain correlative. Convergent associative evidence across disciplines through triangulation will likely be the closest achievable approximation.

Proposed set of criteria for the evaluation of epigenetic inheritance in humans

Given the challenges of epigenetic inheritance research in humans, we propose five interrelated criteria to help collectively evaluate and properly interpret findings in the field (Table 1).

Table 1. Proposed criteria-based framework to evaluate evidence for epigenetic inheritance research in humans

Criteria	Questions for evaluation
I. Exposure definition	Has the environmental exposure been clearly defined and restricted to a time window before conception in the exposed generation? Can generational effects be clearly distinguished?
II. Accounting for confounders	Can observed phenotypes in the offspring be related to epigenetic changes in the exposed parent after accounting for genetic and other confounders (e.g., behavioral or socioenvironmental factors) through adjustment, study design, or moderation analyses?
III. Germline relevance	Are epigenetic modifications present in mature gametes of the exposed parent?
IV. Epigenetic factors	Are corresponding epigenetic changes detected in the offspring and possibly grand-offspring?
V. Functional consequence	Are epigenetic modifications identified in the offspring relevant for phenotypes?

The table outlines five interrelated criteria that should be assessed synergistically. A failure to meet a single criterion should not lead to dismissing the study but should temper conclusions about epigenetic inheritance.

Criterion I: exposure definition

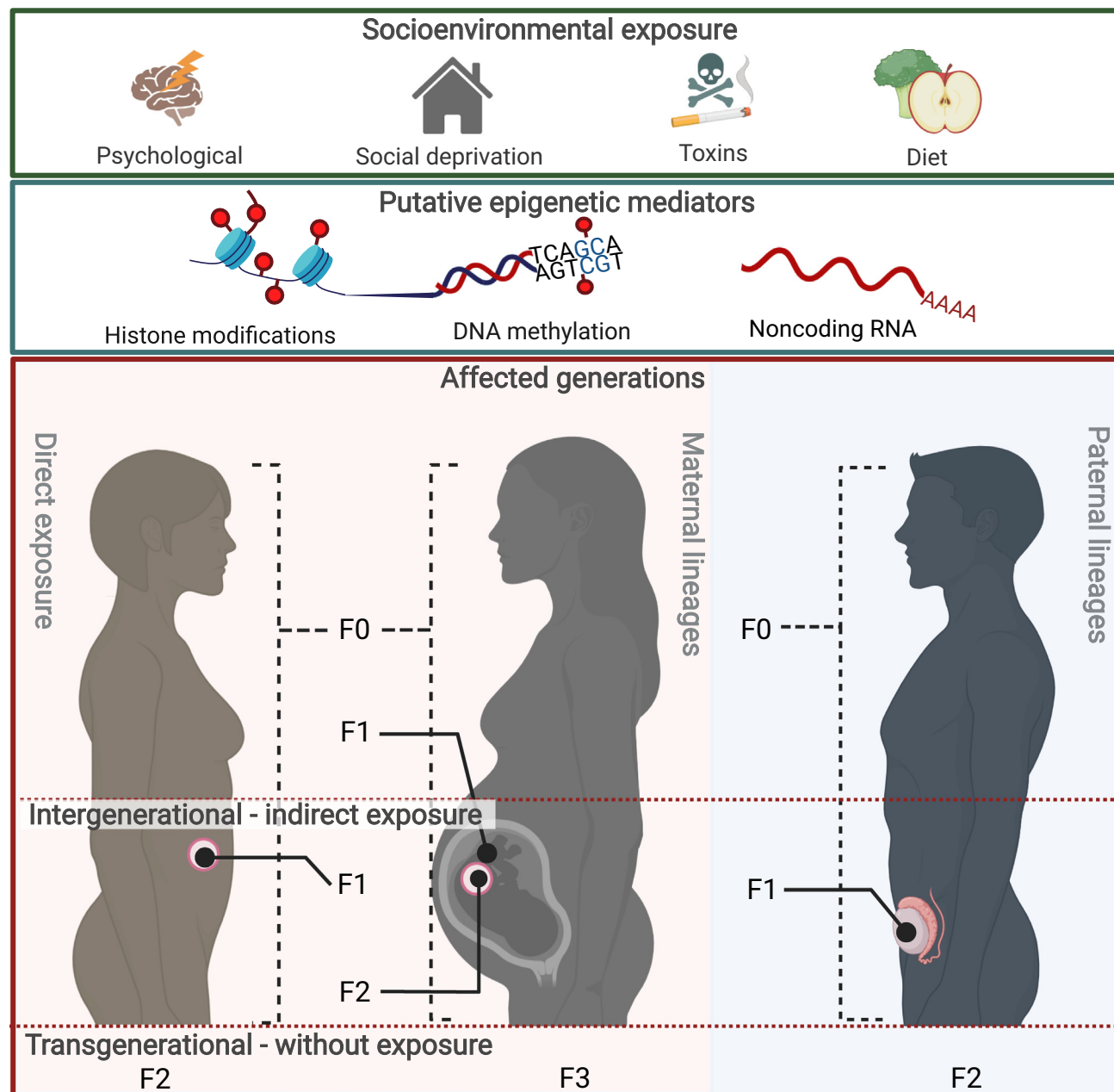
Exposure should be defined as precisely as possible, ideally restricted to an identifiable condition and time window in a single generation, with descendants who are not themselves directly exposed. This is challenging given the inherent heterogeneity and complexity of exposures such as trauma, endocrine disruptors, and poor life conditions; their degree of severity; and their frequency and durability, which can span several life stages. However, it is critical to collect as much information about exposure characteristics as possible across individuals to better delineate their potential consequences. Beyond the exposure itself, the persistence of its effects also matters. While a violent event can be brief, its aftermath may last weeks, months, or years (e.g., post-traumatic stress). In offspring, direct exposure cannot be excluded because the originating germ cells may have been impacted within the body of their exposed parent, whether during fetal, postnatal, or adult life. This means that for the matriline, up to three generations can be affected by exposure during pregnancy: the mother, the fetus, and the fetus's germline (Figure 2). Hence, it is only after the fourth generation, in which no cell has ever been directly exposed, that transmission *per se* can be isolated [72]. Biological samples such as blood, saliva, sperm, and circulating fetal cells, but also, whenever possible, materials such as placenta, umbilical cord blood, and amniotic fluid should be collected to assess markers of exposure (e.g., lipids and corticosteroids) across tissues and generations [73].

Criterion II: accounting for confounders

Human studies have confounding influences that are particularly challenging to control experimentally. Interpreting the effects of famine or trauma across generations is complicated by socioeconomic, socioenvironmental, and biological factors such as income, education, urban–rural status, genetic background, and lifestyle. For historical cohorts, such factors are often poorly known. For existing or new cohorts, regardless of whether they are birth-, inter-, or trans-generational, documenting and accounting for behavioral, socioeconomic, and biological factors is key. Rigorous design-based strategies, including a large number of subjects, family-based comparisons (e.g., sibling inclusion, mother/father–offspring duo, or mother–father–offspring trio designs), negative controls, and timing contrasts based on longitudinal data are recommended [74].

Criterion III: germline relevance

The germline is the principal biological vector for parent-to-offspring information transfer. Epigenetic and RNA changes have been observed in the sperm of men after childhood trauma, reported retrospectively, or after endurance training [75–78]. However, this does not indicate whether such changes are passed to the embryo and, if so, whether they modify offspring phenotypes. Furthermore, changes detected in mature gametes do not imply that the germline itself is affected, as this depends on the timing and severity of exposure. In males, exposure in childhood (before spermatogenesis) can affect progenitor cells in the testes, which may cause changes to become ingrained in the germline and to persist across life, affecting all produced spermatozoa. Exposure in adulthood can affect sperm cells only during maturation in the epididymis (outside the testes) and not be ingrained in progenitor cells, leaving the germline intact. This has distinct implications for the stability, transmission, and functional impact of changes in the offspring. Somatic cells in the gonads may also be affected by exposure and contribute to changes in gametes, for instance, via nutritional factors [79]. Although limited by ethical and methodological constraints, collecting and analyzing human gametes is an important step in documenting the effects of exposure. Whenever possible, analyses in children should also be conducted to assess correlations. Causality between exposure, germline, and offspring phenotypes can be investigated in rodent models, providing experimental support for findings in human studies.



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Figure 2. Biological representation of inter- and trans-generational effects of selected socioenvironmental exposures. The figure illustrates how selected socioenvironmental exposures can affect several generations directly and how to consider such effects in order to distinguish direct exposure from transmission. Upon exposure, maternal (left) or paternal (right) F0 somatic and reproductive cells can be directly affected. Reproductive cells in mothers are oocytes in the ovaries or the uterus and, in fathers, spermatogenic cells (spermatogonial stem cells, spermatocytes, and spermatids) in the testis and spermatozoa in the epididymis. These cells will give rise to the offspring (F1) upon reproduction, and if they carry epigenetic changes due to exposure, these changes will thus be part of the offspring. This, therefore, represents direct exposure at the germ-cell stage in the exposed parent (with no transmission involved). In a pregnant mother, the fetus (offspring, F1) and the embryonic germ cells in the fetus (grand-offspring, F2) are directly exposed. Germline-dependent transmission can only be inferred when neither the individual nor the germ cell from which it originated has been directly exposed. This figure was created using BioRender. Løkhammer, S. (2026).

Criterion IV: epigenetic factors

Determining which factors to analyze is challenging, as the epigenome and transcriptome are complex and difficult to relate to a specific exposure or phenotype. Whenever possible, several factors and RNA should be examined concurrently in the same tissues or cells and in large numbers of parents and offspring, using well-established methods to ensure reproducibility and reliability. Importantly, changes are not expected to be the same in parents and children because of different temporal and age-dependent contexts. In parents, changes are induced directly at the time of exposure and/or thereafter, while in offspring they can be present from the very beginning in the originating gamete and persist through development and adulthood.

Criterion V: functional consequences

Demonstrating that environmentally induced epigenetic changes influence gene expression and ultimately contribute to phenotypic traits is an important but difficult step. A pragmatic approach is to focus on candidate epigenetic marks or RNAs that can be mechanistically related to the observed traits, such as 'PPAR α ' regulators for metabolic dysregulation or the glucocorticoid receptor gene *NR3C1* for stress-related behaviors. In animal models, causality can be tested by mimicking epigenetic changes using clustered regularly interspaced short palindromic repeats (CRISPR)-based epigenome editing tools in germ cells or embryos, or by microinjecting candidate RNA into control zygotes and assessing the effects on gene expression and phenotypes. However, complex, genome-wide epigenetic and RNA profiles cannot be reproduced; thus, causality can be tested only on candidate loci. Furthermore, most phenotypes (e.g., depression or metabolic syndrome) are polygenic, environmentally modulated, and slowly developing (over decades), and cannot be attributed to a few epigenetic marks or RNAs.

Implications and further research directions

The field of epigenetic inheritance in humans stands at a conceptual and methodological inflection point. Current evidence is correlational, tissue-restricted, or cross-sectional, and thus insufficient to resolve whether environmentally induced epigenetic changes in the germline causally shape offspring phenotypes. Advancing the field requires a fundamental and rigorous rethinking of what constitutes meaningful evidence and experimental design.

A critical unmet need is large-scale integration of epigenomics with genetics. Most human epigenetic studies remain largely confined to DNAm in peripheral blood or blood cells. This is only one epigenetic modification among many, and blood is of limited relevance to the germline. Moreover, these approaches rarely model genetic and epigenetic effects jointly. The omnigenic model, in which complex traits reflect the cumulative effects of many genetic variants across regulatory networks, suggests that familial clustering—often attributed to epigenetic inheritance—is more parsimoniously explained by shared polygenic risk and exposures than by germline epigenetic transmission [80]. Disentangling these contributions can be attempted through genome-wide genetic and epigenetic profiling in the same individuals across generations, and through statistical frameworks (e.g., Mendelian randomization and structural equation modeling).

Technological advances, including long-read sequencing platforms that enable simultaneous profiling of DNAm, chromatin accessibility, and genetic variants at single-molecule resolution in individual cells, combined with spatial multi-omics, will help characterize germline epigenomes with high precision and sensitivity. These tools may allow us to determine whether exposure-induced marks are present in human embryos—an important question to answer. Additionally, longitudinal, multigenerational cohorts with repeated biospecimen collection, comprehensive exposure histories, and deep phenotyping can provide essential epidemiological infrastructure for such work.

The medical and societal implications of this research are considerable. Integrating parental and grandparental exposures as modifiable risk factors can reframe disease prevention through periconceptional interventions (e.g., nutritional, psychological, or pharmacological) that may reduce heritable risk. It also raises profound ethical questions regarding responsibility, stigmatization, screening practices, and the use of epigenetic data in insurance, reproduction, and public health policy. This will require developing robust legal and ethical frameworks alongside scientific progress (see [Outstanding questions](#)).

Concluding remarks

Although biologically plausible, epigenetic inheritance in humans remains unproven, and existing evidence is largely correlational. Establishing causality is challenging due to multiple confounders, genetic complexity, difficulties in accessing relevant tissues, and limited experimental designs. Mechanistic evidence from model organisms is strong—the germline epigenome can support transmission—and epidemiological data are consistent with this possibility, yet neither establishes causality in humans. The central challenge is defining rigorous criteria to distinguish germline-mediated transmission from genetic, social, behavioral, and *in utero* confounding, to complement correlational observations.

The framework proposed here is intended as an evolving guide to be updated as methods improve and biological advances accumulate. The outlined criteria do not establish causality but provide a structured approach to assess the plausibility of epigenetic inheritance claims. Progress requires acknowledging current limitations, as overinterpretation undermines credibility and misdirects research, whereas outright dismissal overlooks substantial experimental evidence from model systems. Going forward, studies should integrate genetic and epigenetic variation, incorporate multiple molecular layers, and adopt longitudinal, multigenerational designs with adequate controls for causal inference. Resolving the contribution of epigenetic inheritance to human traits will refine our understanding of heredity, development, and disease.

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Declaration of interests

The authors declare no competing interests.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly AI in order to review the manuscript for grammar and identify relevant bibliographic references. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Outstanding questions

Which different types of—and how many—epigenetic changes need to occur together in a sperm or egg cell to actually affect the health or behavior of the next generation?

How do the age at which a parent is exposed and the duration of exposure determine if the biological effects can be passed on to their children—and does this differ between mothers and fathers?

What analyses and methodological strategies can help identify confounding factors such as *in utero* effects, shared environment, cultural transmission, or parental behaviors in studies of epigenetic inheritance?

How can cross-disciplinary methods integrating epidemiology, molecular biology, and social sciences improve research on epigenetic inheritance in humans?

Can advances in computational modeling enhance the ability to distinguish genetic from epigenetic contributions to heritable traits?

Can improved experimental paradigms or natural experiments (e.g., population bottlenecks and adoption studies) provide novel insight into epigenetic inheritance?

Can advances in imaging or *in vivo* tracking enable direct observation of epigenetic modifications in germ cells or early embryos?

What frameworks can optimize accurate communication of findings and uncertainties in human epigenetic inheritance?

How can emerging findings in this field influence ethical considerations, public health policy, and risk communication?

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